

Simulation and Analysis of Customer Queuing System in a Bank Using R

Scenario

A well-established bank is experiencing high customer footfall, leading to long queues and increased waiting

times. The bank management wants to optimize the service process by simulating the current queuing system

and predicting future performance under different scenarios.

Problem Statement

The bank operates with:

- Two service counters, each with a single server.
- Customer arrivals follow a Poisson distribution with an average rate of 10 customers per hour.
- Service times following an Exponential distribution with an average service time of 5 minutes per customer.
- Customers are served on a first-come, first-served (FCFS) basis.
- The bank manager is considering hiring an additional counter if the average waiting time exceeds 15 minutes.

Tasks

1. Model Development

- Formulate the bank's queuing system using appropriate probabilistic models.
- Implement the simulation in R, using suitable packages such as queueing or custom functions.
- Simulate the system for an 8-hour workday (480 minutes).

2. Analysis and Interpretation

- Analyze the performance of the current system, calculating key performance metrics such as:
 - Average queue length
 - Average waiting time
 - Server utilization
- Evaluate the impact of adding counter.
- Discuss the implications of the findings for the bank's operational strategy.

3. Visualization and Reporting

- Provide appropriate visualizations such as histograms and line charts.
- Prepare a concise report summarizing the methodology, results, and recommendations.